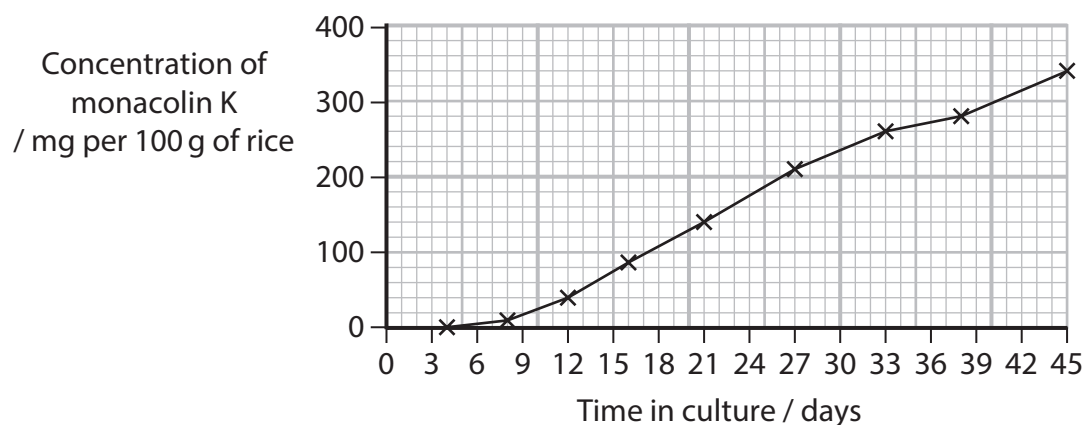


7 Statins are used in the prevention and treatment of cardiovascular disease (CVD).

Red yeast rice (RYR) is a source of food and medicine in Asia. It is produced by the fermentation of rice by yeast.

Red yeast rice contains monacolin K, which is similar to a statin.

- (a) The graph shows the production of monacolin K during the fermentation of rice by yeast.



Calculate the mean rate of production of monacolin K in this culture, in the 45-day time period.

Give your answer to **two** decimal places.

(2)

Answer mg per 100 g of rice per day

*(b) Scientists have studied the effects of RYR on lipid levels.

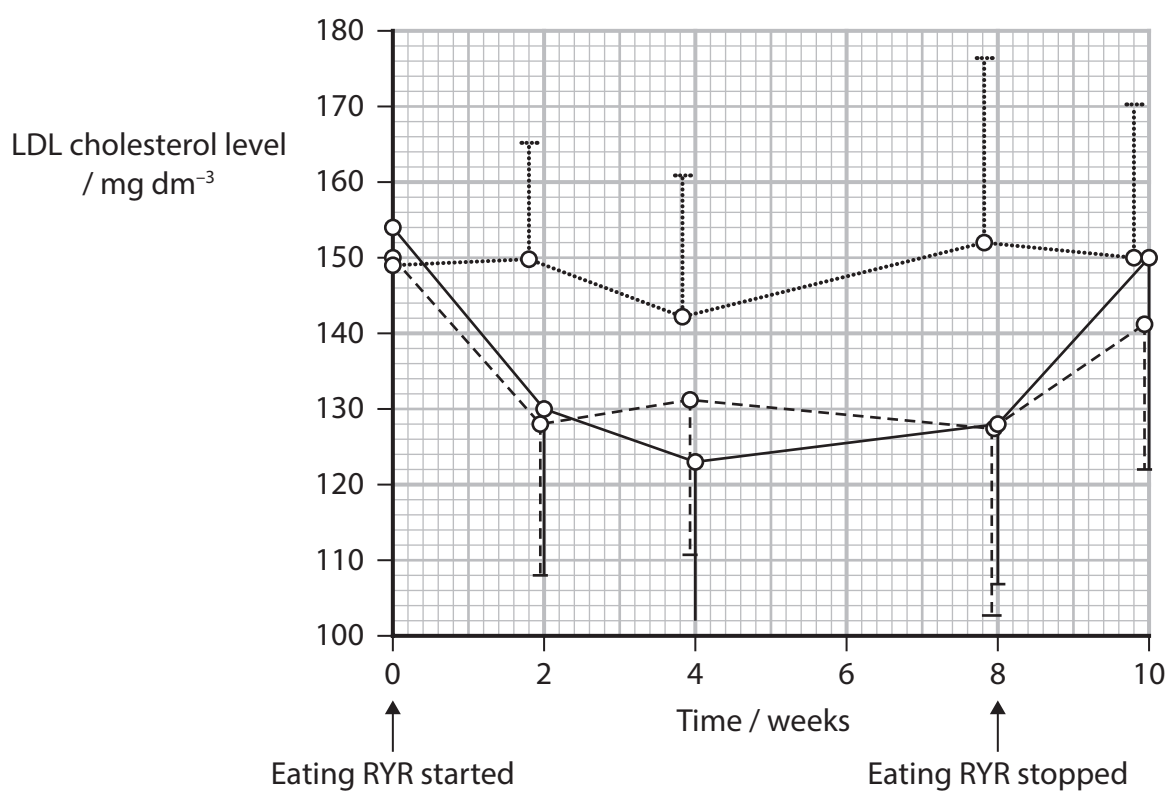
The table shows the results of four studies on the effects of eating RYR on lipid levels.

Study	Mass of RYR / mg per day	Percentage change in lipid levels (%)		
		Decrease in total cholesterol	Decrease in LDL cholesterol	Change in HDL cholesterol
1	340	16 to 17	21 to 24	Between a decrease of 3 and an increase of 3
2	1200	20	34	Increase of 18
3	2400	13	23	No data
4	1200	13	23	No data
	2400	24	32	No data

Graph 1 shows the effects of eating RYR daily on LDL cholesterol levels.

It also shows the changes in levels of LDL cholesterol when eating RYR was stopped.

The investigation included a control group that did not eat RYR.



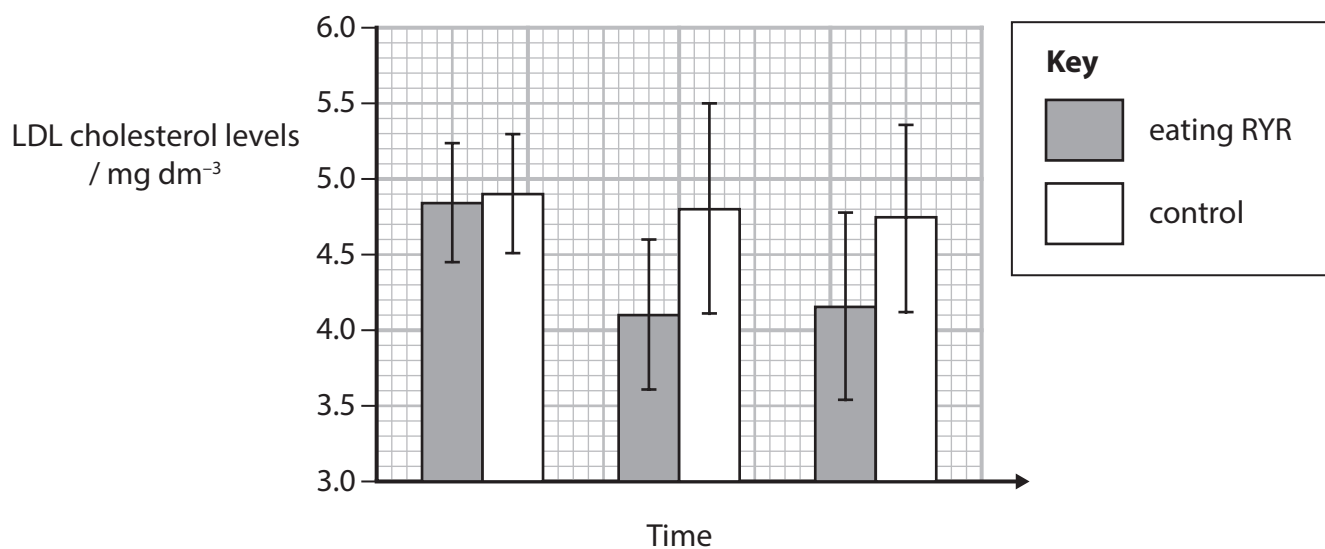
Key

..... control (n = 22) — eating 200 mg RYR (n = 21) --- eating 100 mg RYR (n = 22)

Graph 1



Graph 2 shows the results of another study where the effect of eating RYR on LDL cholesterol levels was compared with a control group.



Graph 2

Discuss the limitations of using the results of these studies to draw conclusions about the effects of RYR on lipid levels.

(6)



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(c) Treatment with statins can have side effects.

(i) Name **two** possible side effects of being treated with statins.

(1)

1

2

(ii) Describe how an investigation could be designed to identify any side effects of treatment with monacolin K.

(4)



- (iii) Suggest **two** problems when assessing possible side effects resulting from treatment with monacolin K in an investigation.

(2)

1

.....

.....

2

.....

.....

(Total for Question 7 = 15 marks)

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